

EVIDENCE ROOM

PROFESSIONAL · AGENT LIBRARY

A03 — Matching Agent

FIELD	VALUE
Agent ID	A03
Name	Matching Agent (2-way / 3-way)
Domain	Invoice-to-pay / AP
Forrester (Mar 2025) map	Matching
Starting autonomy	Level 1
Human owner (role)	AP Matching Lead
Backup owner (role)	AP Quality Lead
Dispatcher	A16
Product	Evidence Room AP Agent OS · Pro
Charter version	1.0 · September 2026

1. Purpose

Propose a 2-way or 3-way match of each validated PO invoice to purchase-order and goods-receipt lines, apply the *written* tolerance table, and either recommend match-success or a coded variance — without posting, paying, or inventing tolerance.

2. Job description

A03 is the match engine. Forrester (March 2025) names matching as a core AI AP use case. This charter is the control wrapper around that use case: which documents, which documents need a GR, which pennies may pass, and who overrides.

In population: PO invoices and credit notes that A02 passed to match. Multi-line, multi-PO, service-entry, and blanket-release invoices are in.

Out of population: true non-PO (A07), statements (A11), invoices with open DUP-SUS, and invoices whose PO is not in the same ledger.

2-way: invoice ↔ PO (typically services with SES / receipt-not-required).

3-way: invoice ↔ PO ↔ GR / SES.

The policy table — not the model — decides which PO types require a GR. A05 is called when the GR is the missing piece.

Done at Level 1: a match proposal the matching team accepts, edits, or rejects. **Done at Level 0:** shadow match vs historical human matches; no queue recommendations.

Ardent Partners (2024) reports Best-in-Class exception rates of 9% and 59% lower exceptions than peers. A03's job is to *apply policy*, not to promise those rates.

3. Inputs / required data

DATA	REQUIRED?	SOURCE	IF MISSING
Validated work object	Yes	A02	Return to A02
Invoice lines	Yes	A01 extract (human-accepted)	LINE-ERR
PO header + lines (qty, price, account, vendor, remaining)	Yes	ERP	P0-MISS
GR / SES lines where policy requires 3-way	Yes if 3-way	ERP	GR-MISS → A05
Tolerance table version	Yes	Policy store	Hard stop. Do not match.
Match policy (2-way vs 3-way by PO type / item category)	Yes	Policy store	Hard stop
Unit-of-measure conversion	Yes if UoM differs	Material master	QTY-VAR or P0-QLTY
Prior allocations on the PO	Yes	ERP	Risk of over-match
Envelope (if Level 3+)	Yes at those levels	Register	Drop to Level 1

4. Tools / systems

FUNCTION	SAP S / 4HANA	D365 F&O / BC	ORACLE FUSION	NETSUITE	WORKDAY	OTHER
PO	EKKO / EKPO	Purch table	PO	Purchase order	Purchase order	P2P suite
GR / SES	MSEG / ML81N	Product receipt / product receipt journal	Receiving / SES	Item receipt	Receipt	WMS
Match / park	MIRO	Invoice matching	Payables match	Vendor bill match	Invoice match	
Tolerances	OMR6 / T169G	Matching policy	Matching rules	Bill matching prefs	Match rules	Policy store
GR/IR	WRX / F. 13	Accrual	Accrual	Accrual	Accrual	A13 reads

Read: PO, GR, tolerances, open allocations.

Write at Level 1: match proposal on the work object.

Write at Level 2: parked match allocation.

Write at Level 3+: post *only* inside envelope.

Write-never: payment, tolerance-table edit, PO price change, GR create (A05 chases; warehouse posts).

5. Responsibilities

1. Load match policy and tolerance table versions onto the object before any line pairing.
2. Pair invoice lines to PO lines (explicit PO/line, then material + price, then residual qty). Record the pairing method.
3. Decide 2-way vs 3-way from policy, not from "GR exists".
4. Compute qty, price, and amount variances per line and in header total.
5. Apply tolerances **as written**. Inside → match_ok. Outside → QTY-VAR / PRC-VAR.
Missing GR → GR-MISS. Unusable PO → PO-QLTY to A06.
6. Handle multi-line and multi-GR: propose allocation; do not collapse lines to hide a variance.
7. Handle partials and subsequent invoices against the same PO remaining qty.

8. Recommend `match_ok`, `match_ok_with_tolerance` (show the cents), or exception with codes.
 9. At Level 1, wait for human decision. Do not post.
 10. Pass exceptions to A04 with evidence (PO line, GR doc, calculations).
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6. Explicit exclusions

- Never release a payment, bank file, or positive-pay file.
 - Never conclude fraud.
 - Never grant itself a higher autonomy level.
 - Never invent a tolerance, tax position, or write-off.
 - Never use real client/employer data in shipped examples or prompts.
 - Never post outside the envelope or at Level 0–2.
 - Never create a GR to “complete” a 3-way match.
 - Never change PO price or qty.
 - Never net a price variance against an unrelated line.
 - Never match a SAP PO to a NetSuite bill.
 - Never treat header-only match as line match when policy says line-level.
 - Never clear DUP-SUS by matching.
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7. Human owner

AP Matching Lead owns pairing quality, tolerance application, and the variance queues that are still AP’s (vs buyer or warehouse). Backup: **AP Quality Lead**.

Procurement owns PO content (A06). Warehouse owns GR posting (A05). Matching owns the arithmetic and the policy application.

8. Approval requirements by autonomy level

LEVEL	AGENT MAY	HUMAN MUST APPROVE	PROOF ARTEFACT
0 Observe	Shadow match vs human	Lead reviews shadow deltas weekly	A03_shadow_pack
1 Recommend		Accept / edit / reject each	match_decision

LEVEL	AGENT MAY	HUMAN MUST APPROVE	PROOF ARTEFACT
	Propose match / exception		
2 Prepare	Park allocation on the invoice	Human posts	Parked MIRO / bill
3 Execute within guardrails	Post match_ok inside envelope	Sample + all variances	Envelope + sample
4 Managed autonomy	Post named population	Recertify; exceptions only	Register + IA ack

match_ok_with_tolerance at Level 3 is allowed only if the envelope cites the tolerance table version. Payment is still human (A12).

9. Escalation criteria

TRIGGER	ESCALATE TO	CODE
Variance above <i>exception</i> threshold (Northline illustrative: \$500 or 5%, whichever is stricter, policy — not invented)	Buyer via A09 + A04	PRC-VAR / QTY-VAR
No GR after policy wait (illustrative: 3 operating days)	A05	GR-MISS
PO vendor ≠ invoice vendor	A06 + A08	PO-QLTY
Remaining PO qty insufficient	A06 / buyer	QTY-VAR
Multi-PO invoice cannot be allocated without residual > tolerance	Matching Lead	LINE-ERR
Tolerance table unpublished	A16 hard stop	CTRL-BRK
Shadow vs human disagreement cluster on one material	A15 + Procurement	process

Do not escalate: variances inside tolerance. Do not escalate freight billed on a planned condition if the PO has that condition — allocate it.

10. Output standard

10.1 match_proposal

```

policy_version
tolerance_version
match_type          # two_way | three_way

```

```

pairing_method          # explicit | material | residual
lines[].invoice_line
lines[].po_id
lines[].po_line
lines[].gr_docs[]
lines[].qty_inv
lines[].qty_po
lines[].qty_gr
lines[].price_inv
lines[].price_po
lines[].var_qty
lines[].var_price
lines[].var_amt
lines[].within_tolerance
header.var_amt
result                  # match_ok | match_ok_with_tolerance | exception
reason_codes[]
calc_evidence_uri

```

Done (Level 1): match_decision recorded.

10.2 Shadow pack A03_shadow_pack (Level 0 and ongoing sample)

Line-level comparison of agent result vs human result on a held-out set.

10.3 Tolerance cite

Every match_ok_with_tolerance stores the row applied (absolute, percent, and which won).

11. Control requirements

1. **No unpublished tolerance.** Hard stop.
 2. **Line-level evidence.** Header-only success is forbidden when policy is line-level.
 3. **SOD.** Matching poster ≠ payment releaser. A03 system user has no payment object.
 4. **No GR creation.** Missing GR is an exception, not a write.
 5. **Cross-ledger isolation.**
 6. **Sample.** 100% of Level-3 posts in week 1 of any envelope; then the register sample rate.
 7. **Change control** on pairing heuristics equal to ERP config.
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12. Audit evidence to retain

EVIDENCE	PULL FROM	RETENTION
Match proposals + calculations	Work log	Local financial-record policy
Tolerance and policy versions applied	Snapshot on object	Same
Human decisions + override class	Work log	Same
Posted document numbers (Level 3+)	Ledger	Same
Shadow packs	A14	Same
Envelope breaks	A16	Same

13. KPIs

FAMILY	KPI	FORMULA	SOURCE	CADENCE	OWNER	PROMOTION
Activity	Match coverage	Proposals / A02 pass_to_match	Work log	Daily	Matching	→ 1.00
Operational	First-pass match	Human-accepted match_ok / PO invoices	Decision log	Weekly	Matching	1 → 2
Operational	Exception rate (local)	Exceptions / PO invoices	Work log	Weekly	Matching	Present beside Ardent 9% as <i>context</i> , not a target warranty
Operational	Time in match p90	Hours	Work log	Weekly	A16	
Financial	Posted out-of-tolerance \$	Sum (target 0)	Incident	Monthly	Controls	Auto demote
Financial	Over-matched remaining qty	Count of PO over-allocations	ERP vs log	Weekly	Matching	
Risk	Override-for-error	Agent_error / proposals	Decision log	Weekly	Controls	Band
Risk	GR invented	Count (target 0)	Audit query	Daily	Controls	Auto Level 0

FAMILY	KPI	FORMULA	SOURCE	CADENCE	OWNER	PROMOTION
Risk	Unauthorised post	Count	A16	Daily	Controls	Auto Level 0

14. Performance history fields

Common fields, plus:

two_way_count
three_way_count
match_ok
match_ok_with_tolerance
qty_var
prc_var
gr_miss
po_qlty
tolerance_version
over_allocations
shadow_disagree

15. Starting autonomy level

Level 1 — Recommend. No posting envelope. First promotion: Level 2 prepare-park, SAP NL10 PO invoices only (Northline pattern). See the worked example in `RESPONSIBILITY_MODEL.md §13`.

16. Failure handling

FAILURE	DETECT	DEGRADE	NOTIFY	RESUME
Tolerance table missing	Version lookup	Hard stop	A16 + Matching	After publish
PO locked / in change	ERP	Wait state; do not residual-match	Matching	Retry
UoM conversion missing	Material master	P0-QLTY	A06	After master fix
Partial GR race	GR appears after proposal	Recalculate; new version	Matching	Same object

FAILURE	DETECT	DEGRADE	NOTIFY	RESUME
ERP timeout	Adapter	No half-allocation	Systems	Retry idempotent
Model proposes netting lines	Guardrail	Reject proposal; CTRL-BRK if it persists	Controls	Human allocation

17. Cost monitoring

COST OBJECT	MEASURE	WHO ACTS
ERP reads per invoice	Adapter	Systems
Human match minutes	Time	Matching Lead
False GR-MISS chases	A05 noise	Warehouse liaison
Rework	Reversals after bad match	Assistant Controller

18. Worked example — Northline Industrial Group

Northline (fictional), 4,200 employees, ~18,000 invoices/month, SAP + NetSuite NPC. Matching lead: Diego Alvarez. Policy: 3-way on materials; 2-way on service POs with SES. Tolerance table NL-TOL-2026-04 (illustrative): \$50 or 2% of line, whichever is lower.

Object. Helion HF-88421, \$14,260.00, PO 4500218831, four material lines.

Level 1. A03 pairs by explicit PO/line. Lines 1–3: qty and price exact; GRs 500218810–12 exist. Line 4: invoice 100 EA @ \$12.40; PO 100 EA @ \$12.00; GR 100 EA. Price var \$40.00 (3.3%). $\$40 < \50 but $3.3\% > 2\%$ → **outside** (stricter rule wins). Result: exception / PRC-VAR on line 4. Header not match_ok.

Diego accepts the exception. A16 → A04 → A09 (buyer). A03 does not bump tolerance to \$50 “because it is close”.

Happy path variant. Line 4 PO price \$12.40. All inside. Result match_ok. Diego accepts; he posts MIRO. Agent did not post.

Illegal. Creating GR 500218813 to cover a short receipt. Matching the same Helion invoice to a NetSuite NPC bill. Posting at Level 1.

Control. Calculation sheet stored. Tolerance version stamped. If Diego overrides to post anyway, class is preference and the \$40 is visible on the monthly pack.

Charter conformance

- [x] Purpose is one sentence and names a boundary
- [x] Job description names population in / out
- [x] Inputs table has required/optional and “if missing”
- [x] Tools table covers SAP, D365, Oracle, NetSuite, Workday, Other, and read/write
- [x] Responsibilities are verbs at the starting level
- [x] Exclusions include payment release, fraud conclusion, self-promotion
- [x] Human owner is a role with a backup
- [x] Levels 0–4 approval table is present and not softer than the responsibility model
- [x] Escalation has hard triggers and a do-not-escalate line
- [x] Outputs are named artefacts with fields
- [x] Controls include SOD, completeness, change control, untrusted input
- [x] Audit evidence is pullable
- [x] KPIs cover activity / operational / financial / risk and name a source
- [x] Performance history includes the common fields
- [x] Starting autonomy is 0 or 1
- [x] Failure handling has detect / degrade / notify / resume
- [x] Cost monitoring is present
- [x] Northline example is fictional and specific
- [x] No guaranteed savings / fraud / compliance / ROI claim
- [x] Statistics cited or labelled illustrative